Further Strategies — problem 14.15

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Let (x, y) be a solution $x^2 + xy - y^2 = 1$ in positive integers. Prove that
(a) $\gcd(x, y) = 1$; (b) if $x = y$ then $x = y = 1$; (c) $x \leq y < 2x$;
(d) $(x + y, x + 2y)$ and $(2x - y, -x + y)$ are also solutions. Construct an infinite sequence of solutions, and prove that they comprise all solutions.